

Non-Uniform Filter Banks-based Minimum Distance to Riemannian Mean Classification in Motor Imagery BCI

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Abstract—The motion intention in the motor imagery brain-computer interface is identified by classifying the event-related de-synchronization (ERD) and event-related synchronization ERS characteristics of sensorimotor rhythm (SMR) in EEG signals. When the subject imagines different limbs or different parts moving, the rhythm components and bandwidth will change, which varies from person to person. How to find the best sensorimotor frequency band of subjects is directly related to the classification accuracy of brain-computer interface. To solve this problem, this paper proposes a Minimum Distance to Riemannian Mean Classification method based on Non-Uniform Filter Banks. During the training phase, the EEG signals are decomposed into multiple different bandwidth signal by using multiple band-pass filters firstly; Then the spatial covariance characteristics of each frequency band signal are computered to be as the feature vectors. These feature vectors will be classified by the MDRM (Minimum Distance to Riemannian Mean) method, and cross validation is employed to obtain the best sensorimotor frequency band. During the test phase, the test signals are filtered by the bandpass filter of the best sensorimotor frequency bands, and the extracted spatial covariance feature vectors will be classified by using the MDRM. Experimental results on the BCI competition IV 2a data sets show that the proposed method is superior to MDRM and FBCSP classification methods.

I. INTRODUCTION

Brain-Computer Interface (BCI) is the establishment of new communication and control channels between the brain and the external environment, independent of peripheral nerves and muscles, allowing them to interact directly [1], [2], [3]. Motor Imagery (MI) is one of BCI paradigms, in which the EEG

signal is spontaneous, does not need external stimulation, and has stable and active sensori-motor rhythm (SMR) signals, so this paradigm has attracted wide attention of researchers [4], [5], [6]. In the process of motor imagery, the EEG signal weakened due to that the movement imagination of the opposite limb will generate the event-related de-synchronization signal ERD (Event-Related De-synchronization), and the event-related synchronization model ERS (Event-Related Synchronization) will be added after the exercise imagination. During experiments, the EEG with 8-12Hz μ rhythm and 18-26Hz β rhythm can also be recorded at the sensorimotor area of the brain [7], [8]. However, the rhythm components and bandwidth of SMR will change due to different subjects and different limbs or parts of imagery, which brings great challenges to decoding of motor imagery BCI.

In the motor imagery BCI system, the covariance of EEG signals contains the spatial characteristic information of the brain during motor imagery. The spatial covariance matrix has a symmetric positive definite matrix form, which is the most common second-order features in EEG signal analysis, such as the common spatial pattern (CSP) algorithm [2]. These traditional algorithms assume that the sample features are in Euclidean space, so they cannot directly classify these covariance features except for some transformations. If the transformations are carried out, the transformed features will lead to the deviation of the sample expression and the distortion of the spatial structure, so the relationship between samples in the symmetric positive definite matrix space cannot be truly described. Compared with Euclidean geometry, Riemannian geometry can directly process the covariance feature matrix and obtain the spatial covariance matrices (SCM), which can better represent the relationship between the covariance matrices and is conducive to improving the classification and recognition accuracy for BCI.

In addition, the CSP algorithm is generally used for two-classification problems. In multi-classification problems, it needs to use strategies such as "one-to-

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many” and ”one-to-one” . However, SCM can directly realize the multi-classification of EEG signals without calculating the projection matrix, so it is more simple, efficient and less computation.

Recently, it is found that the effectiveness of spatial distribution features largely depends on the selection of frequency bands, and the optimal sensory motion frequency band is usually related to specific individuals and motor imagery tasks. Most spatial features often choose a wide frequency band (4-40HZ) for signal processing. This frequency band contains most of the information related to motor imagery, as well as other redundant information. When using CSP algorithm, researchers have considered the selection of frequency band to improve classification performance, such as SBCSP (Sub-Band Common Spatial Pattern) [9], FBCSP (Filter Bank Common Spatial Pattern) [10], SFBCSP (Spark Filter Band Common Spatial Pattern) [11], DFBCSP-FR (Discriminative Filter Bank Common Spatial Pattern Fisher Ratio) , etc. FBCSP uses filter banks to filter EEG signals, extracts CSP features in each frequency band signal, and uses mutual information to select the largest K features in each frequency band for classification. It has achieved the effective results in the BCI competition IV 2a and 2b data sets. From FBCSP, it can be seen that the frequency band selection can effectively suppress the impact of noise, extract useful operational imagination signals, and improve the accuracy of motion intention recognition.

Based on this idea, researchers combined filter banks with SCM and proposed FBRTS (Filter Banks and Riemannian Tangent Space) [12], [13] and other methods. For example, reference [14] directly decomposes the original EEG signal into multiple intrinsic mode components (MIMF) at one time, which is essentially a filter bank to cover μ and β Rhythm, preserve the MIMF of 6-26HZ frequency band to reconstruct the EEG signal. Reference [15] discusses the impact of different bandwidth on classification performance. In the range of 4-40HZ, 2^n ($n=5$) Hz is used as the bandwidth, and step=2 is used as the step size to decompose the EEG signal into multiple subbands, and all subband features are fused for classification. Sub-band division is required when using the filter bank method.

The above methods all adopt uniform subband bandwidth. Whether using all sub-bands or selecting multiple sub-bands to extract features, we will face the problem of bandwidth selection: if the bandwidth is too wide, interference information will be

included; If the bandwidth is too narrow, the EEG information of the motor imagery will be incomplete and the motion recognition accuracy will be affected. Therefore, how to find the effective sensorimotor frequency band of the subject is directly related to the accuracy of brain-computer interface motion intention recognition.

In brain science and neuroscience, EEG is usually divided into δ , α , β and other frequency band rhythms, in which α and β rhythms are closely related to motor imagery. According to the characteristics of α and β bands, this paper proposes a method of non-uniform frequency band division, and then extracts covariance matrices based on Riemannian manifolds on each band, and finds the effective sensory motion band through cross validation, Finally, the manifold based MDRM are used for classification and recognition. Because this frequency band division comes from EEG band division of brain science and neuroscience, the effective sensorimotor frequency band found by this division method is more physiologically explanatory.

II. RIEMANNIAN MANIFOLD

A. Riemannian Manifold Theory

A Riemannian manifold is a differentiable manifold with a Riemannian metric in which there is a tangent space at each point , which can be seen as a Euclidean space [16]. Common Riemannian manifolds in BCI applications include Hermitian manifolds, symmetric positive definite (SPD) manifolds, Stiefel manifolds, and Grassmann manifolds, each with different Riemannian measures [17]. Since the covariance matrices in MI-EEG are a positive definite matrices, the measure of the SPD manifolds is also called the affine invariant Riemann metric. The formula is expressed as

$$\delta(C_i, C_j) = \delta(W^T C_i W, W^T C_j W), \quad (1)$$

where C_i and C_j are the two sample points on the Riemannian manifold (i and j denote the i th and j th samples respectively), $\delta(C_i, C_j)$ is the Riemannian distance between the two samples, W is an any invertible matrix, and the superscript T represents the transposition of the matrix.

Assume that the measured EEG signal generation model is $x = A \cdot s$, where s is the linear mixed signal of the brain dipole source, which is determined by the position of the dipole in the brain and the position of the electrode on the scalp, A is the noise mixing matrix in the process of signal transmission from the

source space to the sensor space, and the two sample points C_i, C_j in the sensor space are

$$C_i = x_i \cdot x_i^T = A^T s_i^T s_i A = A^T S_i A, \quad (2)$$

$$C_j = x_j \cdot x_j^T = A^T s_j^T s_j A = A^T S_j A, \quad (3)$$

where S_i and S_j are the sample points of the source space i and j respectively. According to the affine invariance of Riemann metric, the Riemannian distance formula can be written as

$$\delta(C_i, C_j) = \delta(A^T S_i A, A^T S_j A) = \delta(S_i, S_j) \quad (4)$$

It can be seen that the Riemannian affine invariance metric can keep the Riemannian distance of the sample points unchanged in the source space and the sensor space, independent of the mixed noise matrix A . Compared with the European metric, the Riemannian metric has better generalization ability.

B. Minimum Distance of Riemann Mean

The minimum distance of Riemann mean classification algorithm is calculates the Riemannian distance between the test set sample and the covariance matrix C_K of each category to identify the category with the minimum distance [18].

If C_i and C_j are two points on the Riemannian manifold. The distance between two points is calculated as

$$\delta(C_i, C_j) = \|\log(C_i^{-1} C_j)\|_F = \left[\sum_{i=1}^n \log^2(\lambda_i) \right]^{1/2}, \quad (5)$$

where the λ_i 's are the real strictly positive eigenvalues of $C_i^{-1} C_j$, and $\|\cdot\|_F$ is the Frobenius norm of a matrix. The formula of sample mean covariance matrix in Riemannian space is

$$G(C_1, \dots, C_K) = \arg \min \sum_{i=1}^K \delta_K^2(C, C_i). \quad (6)$$

The geometric mean in the Riemannian manifold has unique existence and cannot be directly solved. It needs to be solved iteratively using the gradient descent method [19]. For the cross-session experiment, the data of one session is used as the training set, and the data of the other session is used as the test set. Due to the strong generalization ability of Riemann metric, the test results will not be affected in cross-session.

III. NON-UNIFORM FILTER BANKS-BASED MINIMUM DISTANCE TO RIEMANNIAN MEAN CLASSIFIER

Non-Uniform Filter Banks-based Minimum Distance to Riemannian Mean (NUFB-MDRM) classifier has two important steps: calculation of covariance feature based on nonuniform frequency bands and selection of effective sensory motion frequency bands.

A. Covariance Feature based on Non-Uniform Frequency Bands

The EEG signal X is decomposed into N non-uniform sub-bands within 4-40HZ by band-pass filters. These sub-bands includes α , μ and β three frequency bands, and the band μ and β are divided into high and low parts, which are called high μ , low μ , high β , and low β , and other sub-bands used 4hz and 6hz bandwidth, as well as full frequency band. The advantage of using this method is to include all important frequency bands. The decomposed non-uniform band EEG signal is in the form of $X = [X_1, X_2 \dots X_N]$, and the covariance feature in each band is extracted, and the calculation formula is:

$$C_{i,j} = \frac{1}{T-1} X_{i,j} X_{i,j}^T, \quad (7)$$

where $X_{i,j}$ is the i th frequency band signal after being filtered at the j th trial. superscript T is the transpose of matrix, $C_{i,j} \in R^{n \times n}$ is the covariance matrix at the i th frequency band and the j th trial.

This method not only includes μ and β frequency band range, but also includes part of fixed bandwidth. So it will be reasonable physiological explanation.

B. The effective Sensorimotor Frequency Bands

How to find the effective sensorimotor frequency band of subjects is directly related to the classification accuracy of brain-computer interface. In this paper, the wrapper method is employed to select the effective frequency bands. The basic idea of wrapper method is based on the hold-out method. For each feature subset to be selected, the model is trained once on the training set, and then the feature subset is selected on the test set according to the error size. So when the wrapper method selects a feature subset, the feature subset will be as input to the classifier for training, and it will be selected or not mainly according to the result performance.

In this paper, we take the mean SCM of each non-uniform subband as a feature subset, and function (5) is employed as the objective function. Then the cross validation method is used to obtain the classification

Algorithm 1 Non-Uniform Filter Banks-based Minimum Distance to Riemannian Mean

- 1: Preprocess EEG signals.
 - 2: **if** status = training **then**
 - 3: Multiple non-uniform bandwidth band-pass filters are used to filter EEG signals and divide them into several bands.
 - 4: Extract the sample covariance feature of each frequency band signal in Riemannian space.
 - 5: Calculate the mean covariance matrix of each class.
 - 6: MDRM and cross-validation method are used to obtain the effective sensorimotor frequency band B_{best} .
 - 7: **else**
 - 8: **if** status = testing & $B_{best} \neq \emptyset$ **then**
 - 9: Carry out band-pass filtering of B_{best} for the test dataset.
 - 10: MDRM is used to classify test dataset.
 - 11: **end if**
 - 12: **end if**
-

ability of each frequency band. So it will automatically select the frequency band with the highest score as the effective sensorimotor frequency band of individual specificity, and eliminate the interference of unrelated frequency bands to the maximum.

C. NUFB-MDRM Algorithm

The proposed NUFB-MDRM method has seven steps mainly: (1) preprocess EEG signals, (2) multiple non-uniform band-pass filters are used to filter EEG signals into several non-uniform frequency bands, (3) extract the sample covariance feature of each frequency band signal and transform the sample space from Euclidean space to Riemannian space, (4) calculate the class mean covariance matrix of the sample features in the Riemannian space to obtain the mean points of each class, (5) minimum distance of Riemann mean method and cross validation are used to obtain the effective sensory motion frequency band, (6) carry out band-pass filtering of the effective sensory motion frequency band for the test signal, (7) use MDRM to classify the test signals. The algorithm 1 gives the framework of NUFB-MDRM.

IV. EXPERIMENTS AND RESULTS

A. Dataset and Parameters Setting

The dataset in the experiments come from BCI competition IV 2a dataset [20], this dataset contains

TABLE I
CLASSIFICATION ACCURACY (%) OF MDRM, FBCSP AND
NUFB-MDRM.

subject	MDRM	FBCSP	NUFB-MDRM
S1	78.82	69.80	80.86
S2	46.88	55.50	61.07
S3	70.83	80.50	74.64
S4	61.81	53.81	52.08
S5	50.00	44.40	66.25
S6	47.57	40.27	47.94
S7	66.32	77.43	76.06
S8	75.20	68.40	84.00
S9	74.31	72.50	58.66
AVG	63.49	62.51	66.84

data from 9 healthy subjects who respectively perform four types of motor imagery tasks: left hand, right hand, foot and tongue. Each subject runs two sessions on different dates. The data is recorded by a total of 25 channels, including 22 EEG channels and 3 EOG channels.

In this experiment, the frequency band is divided into seven non-uniform sub-bands: 4-8Hz, 8-12Hz, 12-26Hz, 18-26Hz, 26-30Hz, 30-36Hz, 36-40Hz. The ten-fold cross-validation method is employed to obtain the effective sensorimotor frequency bands.

B. Results

In order to demonstrate the effectiveness of NUFB-MDRM, we compare the performance of MDRM, FBCSP, and NUFB-MDRM three classifiers. The table I shows the classification accuracy of these three methods.

As shown in table I, the classification accuracy of NUFB-MDRM method is better than that of the other two algorithms on subjects 1, 2, 5, 6 and 8. The MDRM is better on subjects 4 and 9, and the FBCSP better on subjects 3 and 7. But the average classification accuracy of the NUFB-MDRM method is 66.84%, which is higher than that of the other two algorithms.

V. DISCUSSION

In order to overcome the frequency difference between individuals and different tasks, this paper proposes NUFB-MDRM method. It can more effectively solve the problems of the variability of the working frequency band and noise interference in the multi-class classification problem. This method combines filter-banks and Riemann method to try

to find the optimal characteristic frequency band for classification. This method has some advantages:

(1) Comprehensive feature information: using the method of frequency band division can avoid the interference of frequency bands unrelated to MI and extract the optimal time-frequency space features.

(2) Higher classification accuracy: compared with wide band feature classification, this method can automatically select the effective sensorimotor frequency bands that subjects have distinctive characteristics, and use the effective frequency bands features to improve classification accuracy.

(3) Stronger generalization ability: because of the affine invariance of Riemannian metric, it can maintain the invariance of feature information between different samples, and only need to calculate the Riemannian distance between the unknown category and the known category to determine the classification and maintain high accuracy.

(4) Frequency bands optimization and stability: sensorimotor rhythm will change non-uniformly with motor imagery. motor imagery EEG is divided into multiple non-uniform frequency bands to maximize the user's optimal frequency bands selection.

VI. CONCLUSION

How to extract the most effective features to recognize multiple motor imagery tasks is a highly relevant and challenging research issue. In order to overcome the frequency difference between individuals and different tasks, this paper proposes NUFB-MDRM method and to solve the problems of the variability of the working frequency band and noise interference in the multi-class classification problem. This method combines filter-banks and Riemann method to try to find the optimal characteristic frequency band for classification. The experimental results on BCI IV dataset 2a show that the average classification accuracy of our method is 66.8%, which is better than other two methods. The method can help the research and development of MI-BCI system.

Although the proposed method has achieved better performance, there are still some aspects that can be further considered. The classification method is very simple, only considering the minimum distance, but not considering the advantages of other machine learning algorithms. In an addition, deep learning technology and transfer learning methods have made great progress in the field of brain-computer interface recently, which will be a new direction worthy of consideration and research in the future.

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